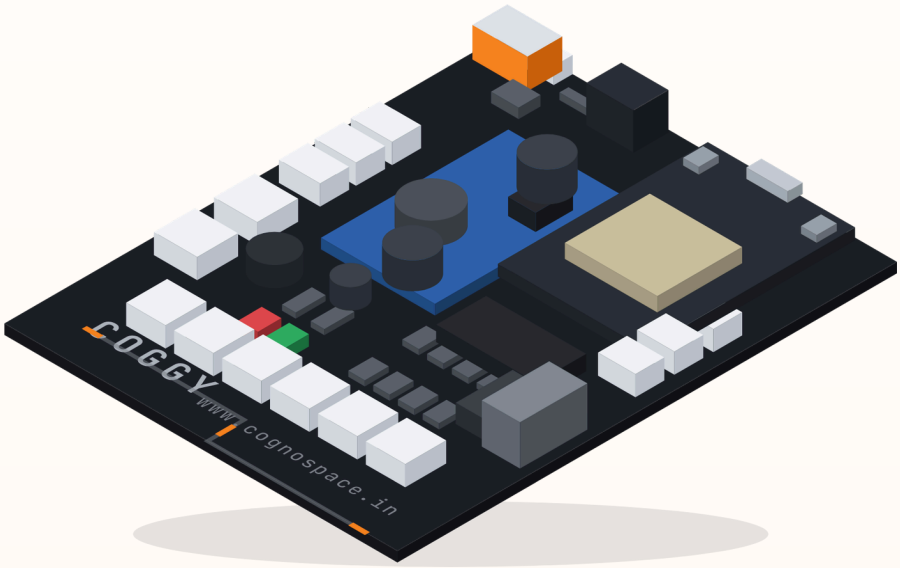
CognoSpace®
Experiential Learning

COGNOSPACE FLASHER · USER MANUAL

Put an activity on your Coggy in four easy steps

This manual shows you how to use the CognoSpace Flasher website. You plug in the Coggy, pick an activity, pick your table and press **Flash**. It takes less than 2 minutes.

4 steps | Under 2 minutes | Chrome or Edge | For teachers & students



What you need

- A **Coggy** board
- A USB **data** cable (some cables only charge)
- A computer with **Google Chrome** or **Microsoft Edge**
- The Flasher website open in that browser

Inside this manual

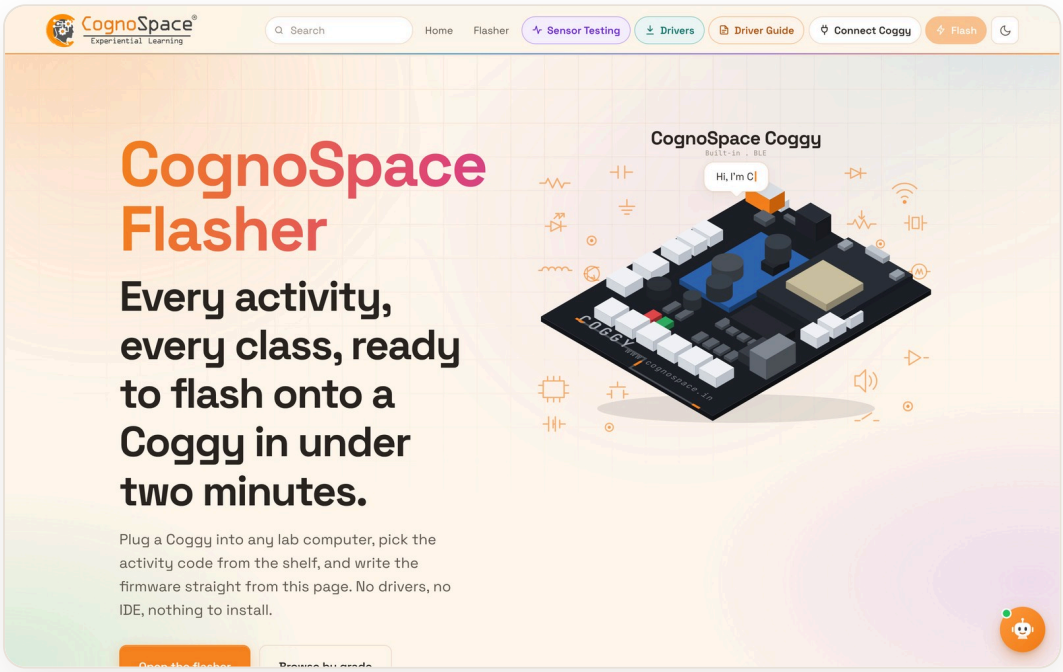
- What the Flasher is, in simple words
- A tour of the screen
- The four steps, with pictures
- Installing the USB drivers
- Wiring sheets, Wi-Fi details and help

GETTING TO KNOW IT

What is the Flasher?

The **Coggy** is CognoSpace's own custom-made circuit board — the **brain** of every activity. It is designed and developed by CognoSpace and powered by an **ESP32** chip. A motor driver, LEDs and sensor pins are already built in, so the same board can run Mechatronics, AI and IoT activities. To do any of them, the Coggy needs the right **program**. The Flasher is a website that sends that program from your computer into the Coggy through a USB cable. Sending the program in is called **flashing**.

Think of it like installing an app on a phone: you choose the app, press install, and the phone can now do something new. Here you choose an **activity** (like *Traffic Signals* or *Robotic Arm*), press **Flash**, and the Coggy is ready for that lesson.



The Flasher's home page. Press **Open the flasher** to start.

Words you will see

Coggy
CognoSpace's ESP32-powered brain board. It runs one activity program at a time.

Flash / Flashing
Writing a program into the Coggy. The old program is replaced by the new one.

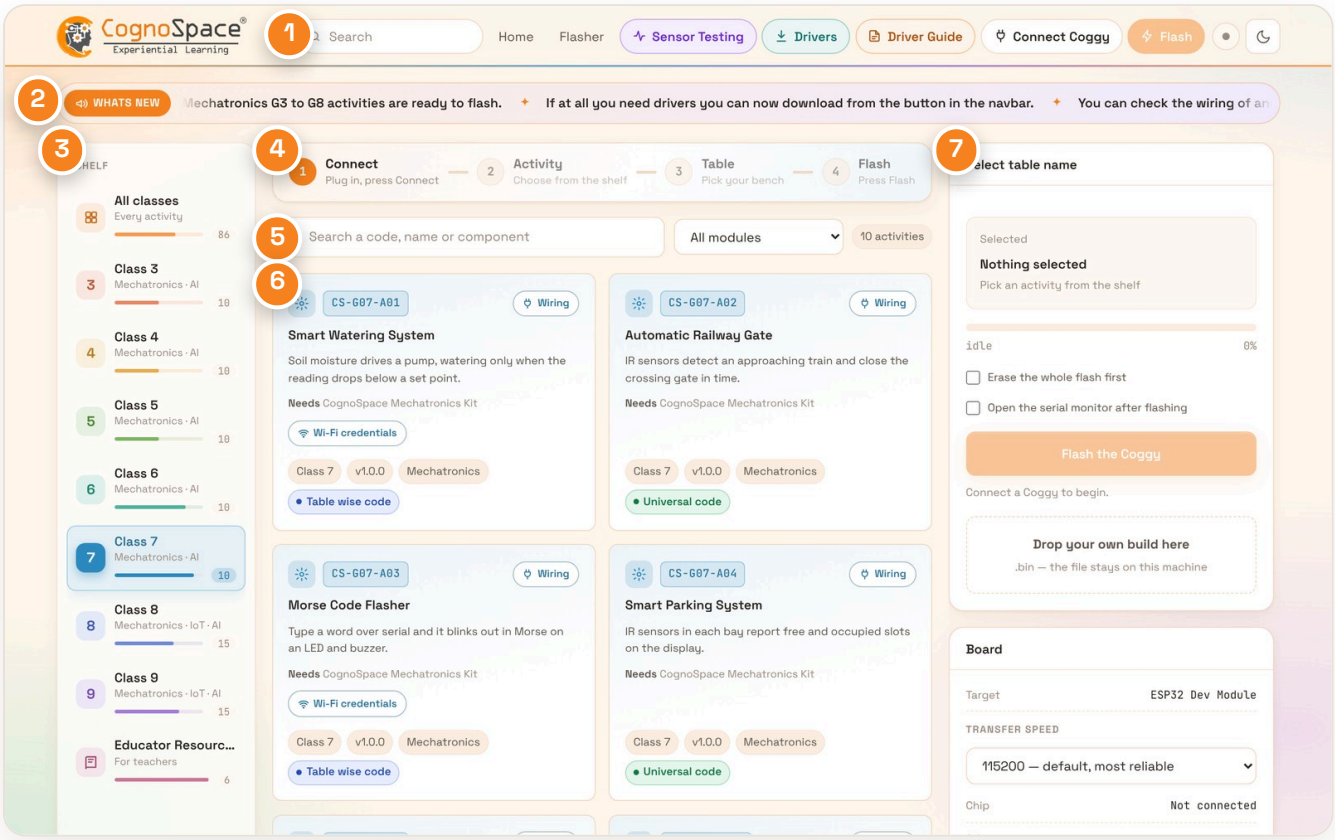
Activity code
Every activity has a short code like `CS-G07-A03`. G07 means Class 7, A03 means activity 3.

Table
Each bench in the lab has a name — T1 Pi, T2 Newton and so on. Some activities need the table so each Coggy gets its own Wi-Fi name.

A QUICK TOUR

The Flasher screen at a glance

After you press **Open the flasher**, this is what you see. Each number below matches a part of the screen.



1 Top bar
Buttons to connect the Coggy, flash it, get drivers and switch to dark mode.

2 What's new
A moving strip with news and tips from CognoSpace.

3 Shelf
Pick your class (3 to 9) or Educator Resources. The cards change to show only that class.

4 Steps bar
Shows where you are: Connect → Activity → Table → Flash. Each done step turns green.

5 Search and filter
Type a name, a code or a part (like "LED"), or choose a module such as AI or IoT.

6 Activity cards
One card for each activity. Click a card to choose it.

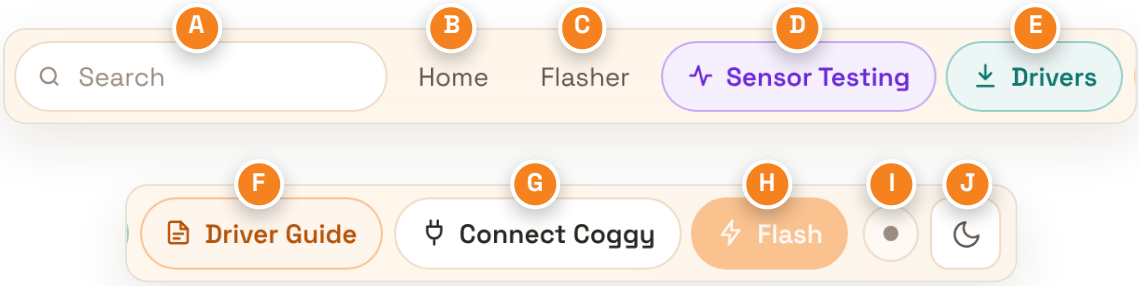
7 Right panel
Shows the activity you chose, the table list, the **Flash the Coggy** button, board details and the console.

? Sparky helper
The round orange robot in the corner answers questions about the Flasher.

THE TOP BAR

What each button does

The bar along the top of the page stays in the same place on every screen.



A — Search
Find any activity by name, code or part.

B — Home
Back to the welcome page.

C — Flasher
Opens the flasher screen with all the activities.

D — Sensor Testing
Opens a separate page to check that sensors work.

E — Drivers
Downloads the USB driver, only needed if your computer can't find the Coggy.

F — Driver Guide
Opens the step-by-step guide for installing the USB driver on Windows.

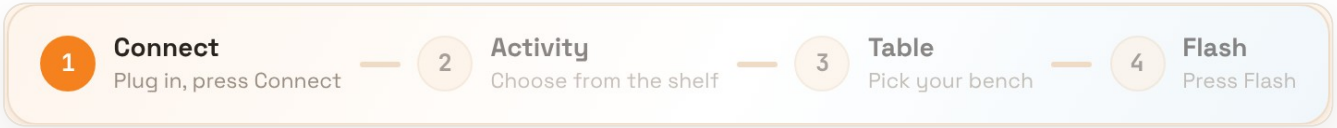
G — Connect Coggy
Links the plugged-in Coggy to the website. After connecting it says **Disconnect**.

H — Flash
Writes the chosen activity onto the Coggy. It lights up once you are ready.

I — Status light
Grey = not connected. Green = Coggy connected.

J — Light / dark mode
Switches the colours of the page. It does not change how anything works.

The steps bar shows your progress



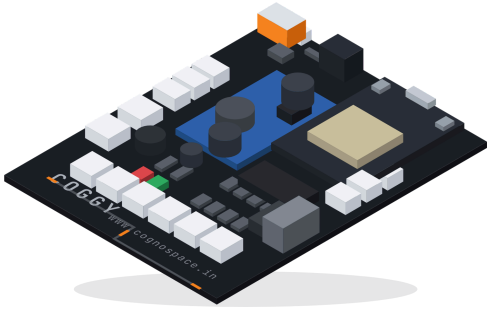
Green tick = done. Orange number = what to do next. Here the Coggy is connected and the activity is chosen; the next step is picking the table.

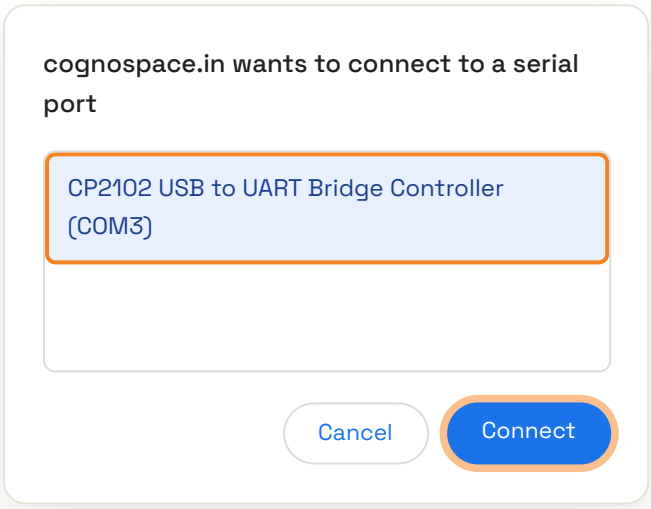
> **Good to know:** the Flasher works in **Google Chrome** and **Microsoft Edge** on a laptop or desktop. It does not work on phones, tablets, Safari or Firefox.

STEP 1 OF 4

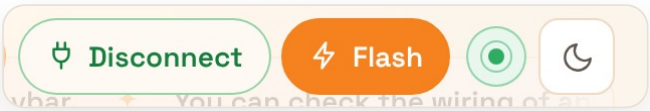
1 Connect the Coggy

1. Plug the Coggy into the computer with a USB **data** cable.
2. On the Flasher, click **Connect Coggy** in the top bar (or **Connect a Coggy** in the right panel).
3. A small box from the browser opens. Click the Coggy in the list, then click **Connect**.





The browser's box. The name may say *CP2102*, *CH340* or *USB Serial*, and the COM number changes from computer to computer.



Connected! The button now says **Disconnect**, the **Flash** button turns orange and the status light turns green.

It worked if the first step in the steps bar shows a green tick and the right panel shows the chip name, for example *ESP32-D0WD-V3*.

The list is empty? Your computer needs the USB driver. See **Install the USB drivers** on the next page. You only do this once per computer.

Tip: if the Coggy doesn't connect, hold the **BOOT** button on the Coggy, click Connect, and let go once the console starts moving.

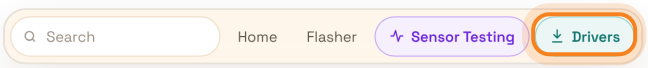
ONLY IF YOU NEED IT

Download and install the USB drivers

Most computers find the Coggy straight away. If the list is empty when you press **Connect Coggy**, or Windows shows the Coggy as an "unknown device", install the drivers once using these steps.

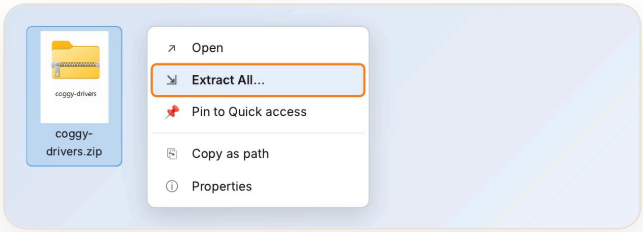
1 Click Drivers in the top bar

The green **Drivers** button downloads `coggy-drivers.zip` straight away.



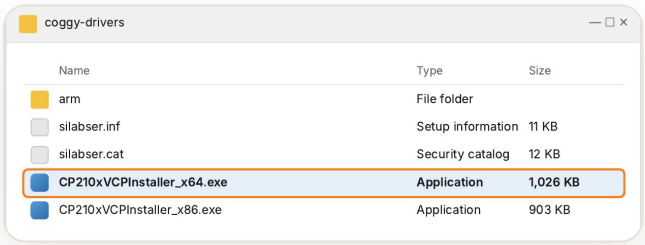
2 Unzip it

Open your **Downloads** folder, right-click `coggy-drivers.zip` and choose **Extract All...**, then **Extract**.



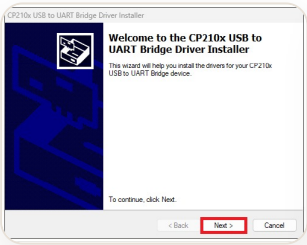
3 Run the installer

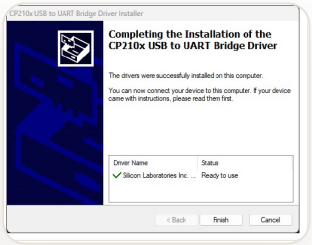
In the new `coggy-drivers` folder, double-click `CP210xVCPInstaller_x64.exe`. Use the `_x86` one only on an old 32-bit PC. Click **Yes** if Windows asks.



4 Click Next, accept, Finish

Click **Next**, choose **I accept this agreement**, click **Next**, then **Finish**.





5 Unplug the Coggy and plug it back in

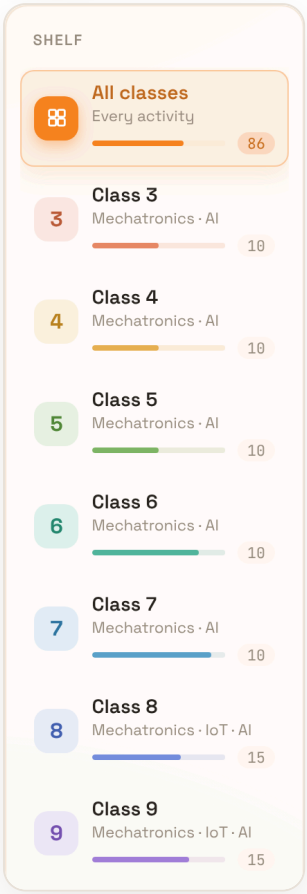
Windows now picks up the driver. Go back to the Flasher and press **Connect Coggy** — the Coggy shows up in the list.

Need more help? Click **Driver Guide** in the top bar for the full guide with every screen, including how to check the Coggy in Device Manager.

Keep the folder. If Windows Update ever replaces the driver, run the installer again from the same folder. You only need to install the drivers once per computer.

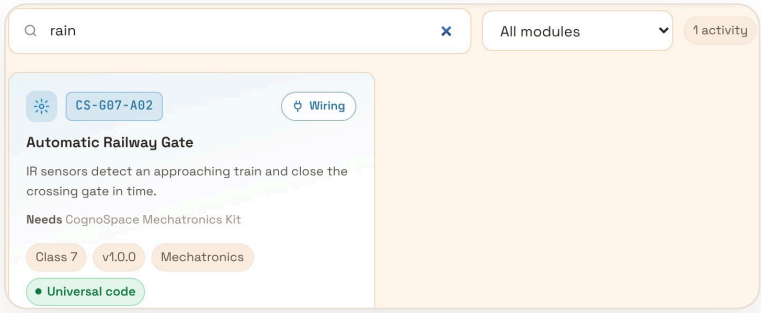
STEP 2 OF 4

2 Choose the activity



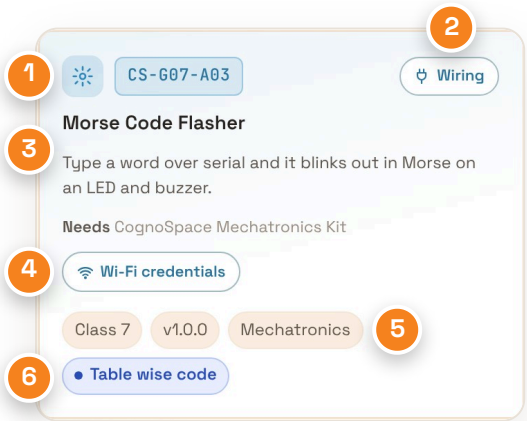
The shelf. Class 7 is chosen.

1. On the **Shelf** on the left, click your **class**. Now only that class's activities are shown.
2. Find the activity your lesson needs. You can also type in the search box — try `rain`, `LED` or a code like `CS-G07-A03`.
3. Click the activity **card**. It gets an orange border and its name appears in the right panel.



Typing "rain" shows every activity that uses rain.

Reading an activity card



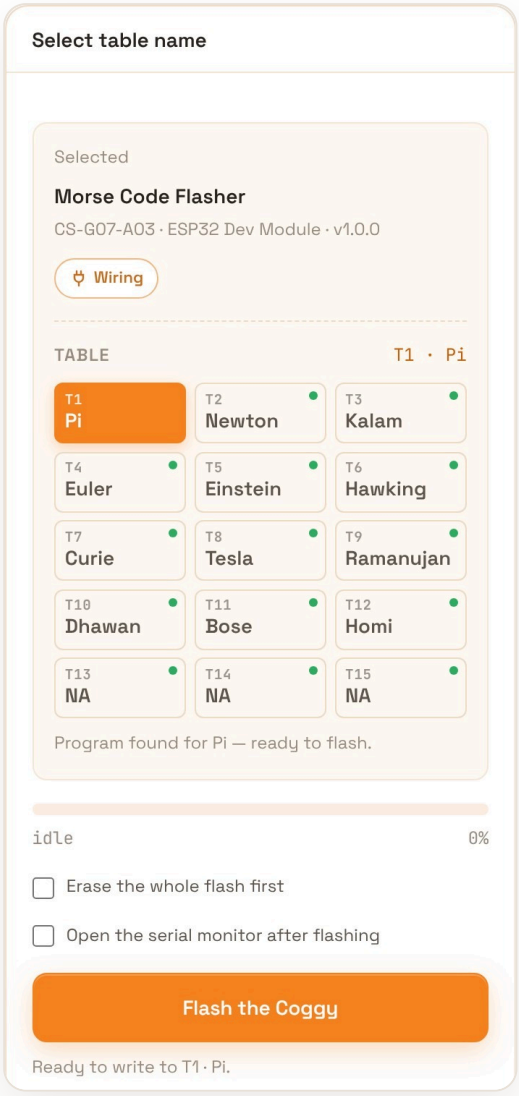
1. **Activity code**
   The short code teachers read out: "flash CS-G07-A03".
2. **Wiring**
   Shows which part goes on which Coggy pin. See page 10.
3. **Name and description**
   What the activity does and the kit it needs.
4. **Wi-Fi credentials**
   Only on activities that make their own Wi-Fi network. See page 11.
5. **Class, version, module**
   Small labels about the activity.
6. **Code type**
   Universal code or Table wise code — explained on the next page.

STEP 3 OF 4

3 Pick your table

Some activities make their **own Wi-Fi network** or Bluetooth name. If every Coggy in the room had the same name, students would join the wrong board. So each table gets its own version of the program.

1. In the right panel, look for **Table**.
2. Click the name of the table this Coggy belongs to, for example **T1 · Pi**. It turns orange.
3. The note under the list says *"Program found for Pi — ready to flash."*



The right panel with **T1 · Pi** picked. The **Flash the Coggy** button is now orange and ready.

Universal code or table wise code?

Universal code

The **same program** works on every table. There is no table list — you go straight to Flash.

Table wise code

Each table has **its own program**. You must pick the table before you can flash.

Lab table names

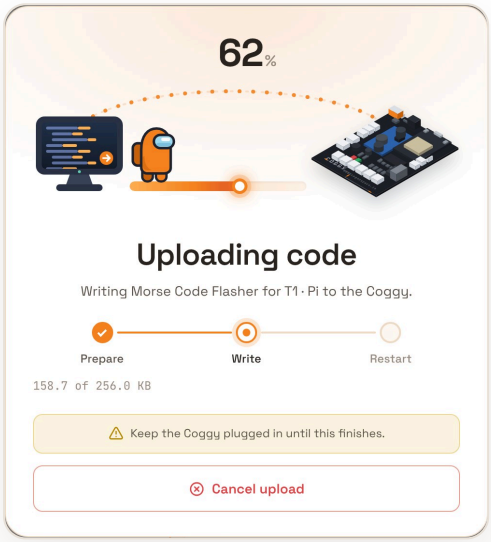
T1 Pi	T2 Newton	T3 Kalam	T4 Euler
T5 Einstein	T6 Hawking	T7 Curie	T8 Tesla
T9 Ramanujan	T10 Dhawan	T11 Bose	T12 Homi

Tip: write the table name on a sticker on each Coggy. Then nobody has to guess which table to pick.

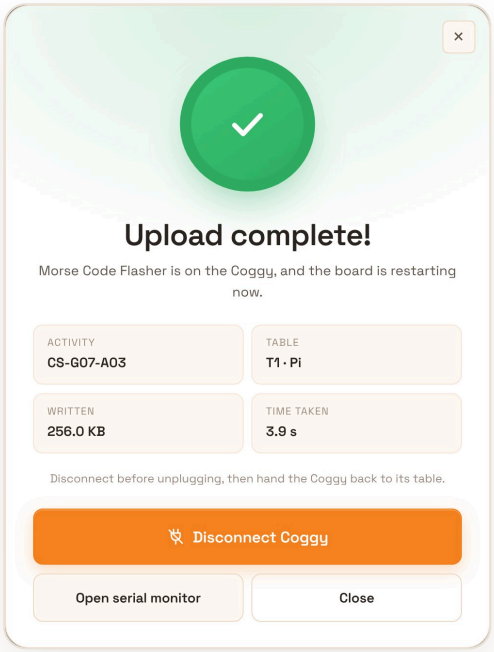
STEP 4 OF 4

4 Press Flash and wait

1. Click **Flash the Coggy** in the right panel (or **Flash** in the top bar).
2. A window opens. Watch the program travel from the computer to the Coggy. It goes through **Prepare → Write → Restart**.
3. When you see **Upload complete!**, the Coggy restarts and starts the activity by itself. The whole upload takes **less than 2 minutes**.



Uploading. The number shows how much is done.



Done! It shows the activity, table, size and time taken.

Don't unplug the Coggy while it says *Uploading code*. If you need to stop, press **Cancel upload** and wait for it to finish stopping.

After it finishes

Disconnect Coggy

Do this before you unplug, then give the Coggy back to its table.

Open serial monitor

See the messages the Coggy prints. Useful for checking it works.

Next Coggy

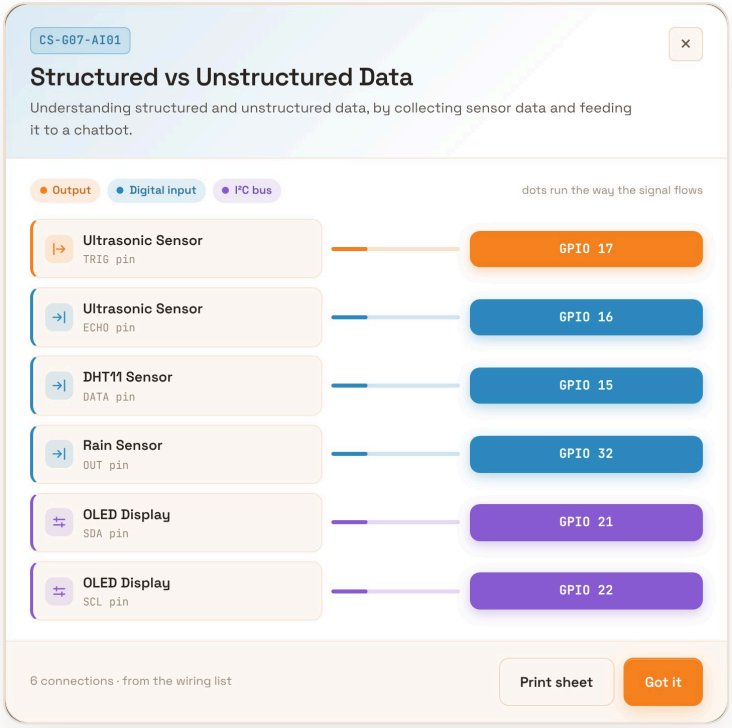
Plug in the next board, press Connect, pick its table and Flash again.

Two optional boxes: **Erase the whole flash first** wipes the Coggy completely before writing — use it only if a board behaves strangely. **Open the serial monitor after flashing** opens the monitor by itself when flashing ends.

FOR THE BENCH

The wiring sheet

Click **Wiring** on any card to see which part connects to which Coggy pin. Show it on the projector, or press **Print sheet** to give one to each table.

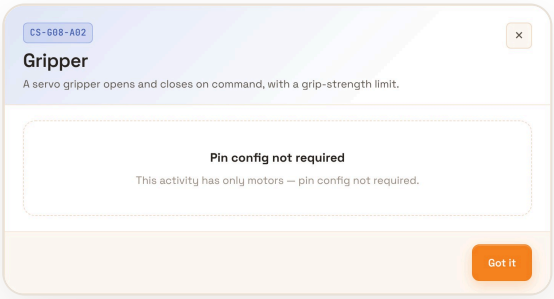


Wiring for *Structured vs Unstructured Data*.

How to read it

- Each row is **one wire**.
- On the left: the **part** and the pin name printed on it, like `TRIG` or `SDA`.
- On the right: the **Coggy pin** to plug it into, like `GPIO 17`.
- Colours show the kind of wire. **Orange** = output, **blue** = input, **purple** = display (I²C).
- **Setup page** means you choose that pin on the activity's own webpage, then wire it to the same pin.

When nothing needs wiring



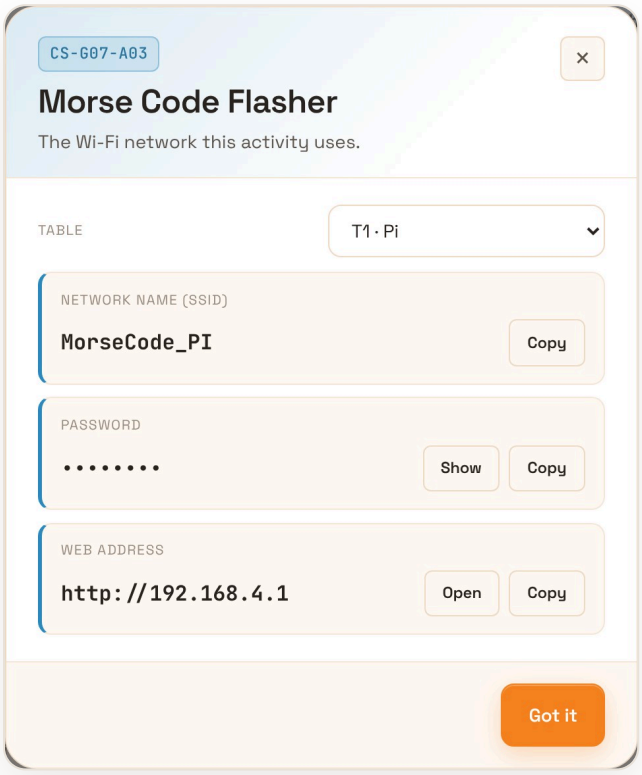
Activities with only motors, or with no parts at all, say **Pin config not required**.

> **Tip:** check the wiring *before* you power the Coggy. A wire on the wrong pin is the most common reason an activity "does nothing".

CONNECTING TO THE ACTIVITY

Wi-Fi credentials

Some activities make their own Wi-Fi network with a webpage on it. Their card has a **Wi-Fi credentials** button that shows the network name, password and web address.



Wi-Fi details for *Morse Code Flasher* on table T1 · Pi.

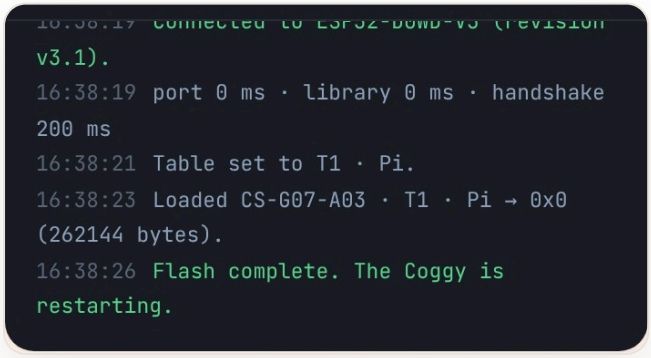
How students join

1. Click **Wi-Fi credentials** on the card and choose the **Table**.
2. On the phone or laptop, open Wi-Fi settings and join the **network name** shown, for example `MorseCode_PI`.
3. Type the **password**. Press **Show** to see it, or **Copy** to copy it.
4. Open a browser and go to the **web address** `http://192.168.4.1`, or press **Open**.

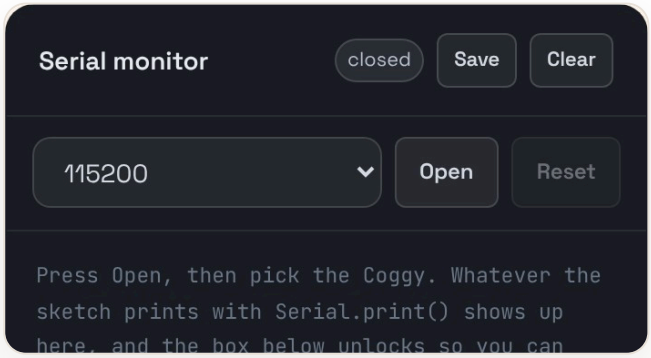
Why the table matters: the end of the network name is the table name. Table T1 gets `…_PI`, table T2 gets `…_NEWTON`. So every bench joins its own Coggy.

Note: the button only shows on table wise activities. Universal code activities don't have it.

The console and the serial monitor



Console — a diary of everything the Flasher did. Green lines mean success.



Serial monitor — press **Open** to read what the Coggy prints, and type to send it messages.

EXTRA FEATURES

More helpful tools

Ask Sparky

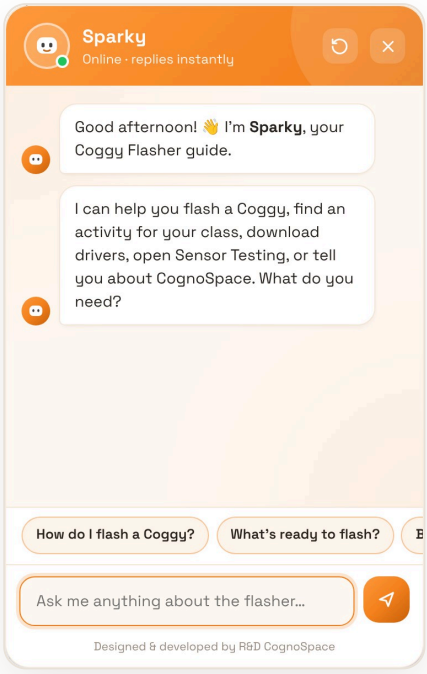
Click the orange robot in the bottom-right corner. Sparky can tell you how to flash, which activities are ready, and where to find the drivers or Sensor Testing. Type a question or tap one of the quick buttons.

Board panel

Under the Flash button, the **Board** panel shows the chip, the USB ID and the board's MAC address after connecting. Leave **Transfer speed** at **115200** — it is the most reliable. Faster speeds only work with a very good cable.

Dark mode

Press the moon button at the end of the top bar. Everything works the same — only the colours change.



Sparky, the Flasher helper.

MEET THE COGGY

CognoSpace's own brain board

The Coggy is a custom PCB **designed and developed by CognoSpace**. It is powered by an **ESP32** chip, so it can think, connect to Wi-Fi and talk over Bluetooth. Everything a class needs is already built onto the board, so students spend their time building and learning — not fixing loose wires.



ESP32 brain
Fast chip with built-in Wi-Fi and Bluetooth.

Built-in motor driver
Runs motors for robots, rovers and arms.

On-board LEDs
Lights to show results straight away.

Ready sensor pins
Plug in sensors, displays and buzzers easily.

Made for CognoSpace activities in Mechatronics AI IoT Robotics

HELP

If something goes wrong

Most problems come from the cable, the browser or the driver. Find your problem below.

“Connect Coggy” is grey

Your browser can’t talk to USB boards. Open the Flasher in **Chrome** or **Edge** on a laptop or desktop.

The port list is empty

Install the USB driver: click **Drivers**, then follow the **Driver Guide**. Also try another cable — some only charge.

It won’t connect

Unplug and plug the Coggy back in. Hold the **BOOT** button, press Connect, and let go when the console moves.

“Port already open”

Close the Arduino IDE or any other serial monitor using the Coggy. Then unplug, plug back in and connect again.

The Flash button stays light

Something is missing. Check the steps bar: connect the Coggy, choose an activity and, for table wise code, pick a table.

Flashing stopped halfway

The cable may be loose or long. Press **Try again**. If it keeps failing, use a shorter cable and keep the speed at 115200.

A table name is not available

The program for that table hasn’t been added yet. Ask the CognoSpace team, or use a table that is ready.

The activity does nothing

Open the **Wiring** sheet and check every wire. Then open the serial monitor to see what the Coggy says.



The whole thing in one line

Plug in → **Connect Coggy** → choose the activity → pick the table → **Flash**. Still stuck? Ask Sparky in the corner of the Flasher, or write to the CognoSpace team at **www.cognospace.in**.